

8.1 General Linear Transformations

Up to now our study of linear transformations has focused on transformations from R^n to R^m . In this section we turn our attention to linear transformations involving general vector spaces, illustrate ways in which they arise, and establish a fundamental relationship between general n -dimensional vector spaces and R^n .

Definitions and Terminology

In Section 1.8 we defined a matrix transformation $T_A : R^n \rightarrow R^m$ as $T_A(\mathbf{x}) = A\mathbf{x}$. We subsequently showed (Theorem 1.8.3) that matrix transformations are precisely the linear transformations from R^n to R^m , i.e., those satisfying

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) \quad \text{and} \quad T(k\mathbf{u}) = kT(\mathbf{u})$$

We use these two properties as the starting point for the following general definition.

Definition 1

If $T : V \rightarrow W$ is a mapping from a vector space V to a vector space W , then T is a **linear transformation** from V to W if for all vectors $\mathbf{u}, \mathbf{v}$ in V and all scalars k :

$$(i) \quad T(k\mathbf{u}) = kT(\mathbf{u}) \quad \text{[Homogeneity property]}$$

$$(ii) \quad T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) \quad \text{[Additivity property]}$$

When $V = W$, the linear transformation T is called a **linear operator** on V .

Combining the two properties shows that for any vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ in V and scalars $k_1, \dots, k_n$,

$$T(k_1\mathbf{v}_1 + k_2\mathbf{v}_2 + \dots + k_n\mathbf{v}_n) = k_1T(\mathbf{v}_1) + k_2T(\mathbf{v}_2) + \dots + k_nT(\mathbf{v}_n) \quad (1)$$

Theorem 8.1.1

If $T : V \rightarrow W$ is a linear transformation, then:

- (a) $T(\mathbf{0}) = \mathbf{0}$.
- (b) $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in V .
- (c) $T(-\mathbf{v}) = -T(\mathbf{v})$ for all $\mathbf{v}$ in V .

Proof. For (a): $T(\mathbf{0}) = T(0\mathbf{u}) = 0T(\mathbf{u}) = \mathbf{0}$. For (b): $T(\mathbf{u} - \mathbf{v}) = T(\mathbf{u} + (-1)\mathbf{v}) = T(\mathbf{u}) + (-1)T(\mathbf{v}) = T(\mathbf{u}) - T(\mathbf{v})$. Part (c) follows by setting $\mathbf{u} = \mathbf{0}$ in (b). ■

EXAMPLE 1 | Matrix Transformations

Every matrix transformation $T_A : R^n \rightarrow R^m$ defined by $T_A(\mathbf{x}) = A\mathbf{x}$ is a linear transformation (since matrix multiplication satisfies both homogeneity and additivity).

EXAMPLE 2 | The Zero Transformation

The mapping $T : V \rightarrow W$ defined by $T(\mathbf{v}) = \mathbf{0}$ for every $\mathbf{v}$ in V is linear (the **zero transformation**), since

$$T(\mathbf{u} + \mathbf{v}) = \mathbf{0} = \mathbf{0} + \mathbf{0} = T(\mathbf{u}) + T(\mathbf{v}) \quad \text{and} \quad T(k\mathbf{u}) = \mathbf{0} = k\mathbf{0} = kT(\mathbf{u})$$

EXAMPLE 3 | The Identity Operator

The mapping $I : V \rightarrow V$ defined by $I(\mathbf{v}) = \mathbf{v}$ is a linear operator on V (the **identity operator**).

EXAMPLE 4 | Dilation and Contraction Operators

If c is any scalar, the operator $T : V \rightarrow V$ defined by $T(\mathbf{x}) = c\mathbf{x}$ is linear, since $T(k\mathbf{u}) = c(k\mathbf{u}) = k(c\mathbf{u}) = kT(\mathbf{u})$ and $T(\mathbf{u} + \mathbf{v}) = c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v} = T(\mathbf{u}) + T(\mathbf{v})$. When $0 < c < 1$ this is called a **contraction** of V with factor c ; when $c > 1$, a **dilation** of V with factor c .

EXAMPLE 5 | A Linear Transformation from P_n to P_{n+1}

Let $\mathbf{p} = p(x) = c_0 + c_1x + \cdots + c_nx^n \in P_n$ and define $T : P_n \rightarrow P_{n+1}$ by $T(\mathbf{p}) = xp(x) = c_0x + c_1x^2 + \cdots + c_nx^{n+1}$. For scalar k and polynomials $\mathbf{p}_1, \mathbf{p}_2 \in P_n$:

$$T(k\mathbf{p}) = x(kp(x)) = k(xp(x)) = kT(\mathbf{p})$$

$$T(\mathbf{p}_1 + \mathbf{p}_2) = x(p_1(x) + p_2(x)) = xp_1(x) + xp_2(x) = T(\mathbf{p}_1) + T(\mathbf{p}_2)$$

so T is linear.

EXAMPLE 6 | A Linear Transformation Using an Inner Product

Let $\mathbf{v}_0$ be any fixed vector in a real inner product space V , and define $T : V \rightarrow R$ by $T(\mathbf{x}) = \langle \mathbf{x}, \mathbf{v}_0 \rangle$. From properties of inner products:

$$T(k\mathbf{u}) = \langle k\mathbf{u}, \mathbf{v}_0 \rangle = k\langle \mathbf{u}, \mathbf{v}_0 \rangle = kT(\mathbf{u})$$

$$T(\mathbf{u} + \mathbf{v}) = \langle \mathbf{u} + \mathbf{v}, \mathbf{v}_0 \rangle = \langle \mathbf{u}, \mathbf{v}_0 \rangle + \langle \mathbf{v}, \mathbf{v}_0 \rangle = T(\mathbf{u}) + T(\mathbf{v})$$

EXAMPLE 7 | Transformations on Matrix Spaces

Let M_{nn} be the space of $n \times n$ matrices. Determine whether the following are linear.

(a) $T_1(A) = A^T$ (b) $T_2(A) = \det(A)$

SOLUTION (a): From properties of the transpose, $T_1(kA) = (kA)^T = kA^T = kT_1(A)$ and $T_1(A + B) = (A + B)^T = A^T + B^T = T_1(A) + T_1(B)$, so T_1 is linear.

SOLUTION (b): $T_2(kA) = \det(kA) = k^n \det(A) = k^n T_2(A)$, which is not $kT_2(A)$ for $n > 1$. Also, $\det(A + B) \neq \det A + \det B$ in general (Section 2.3). So T_2 is not linear.

EXAMPLE 8 | Translation Is Not Linear

If $\mathbf{x}_0$ is a fixed nonzero vector in V , the transformation $T(\mathbf{x}) = \mathbf{x} + \mathbf{x}_0$ is *not* linear, since $T(\mathbf{0}) = \mathbf{x}_0 \neq \mathbf{0}$ (violating part (a) of Theorem 8.1.1). Geometrically, this **translation** shifts each point $\mathbf{x}$ along a line parallel to $\mathbf{x}_0$ through distance $\|\mathbf{x}_0\|$.

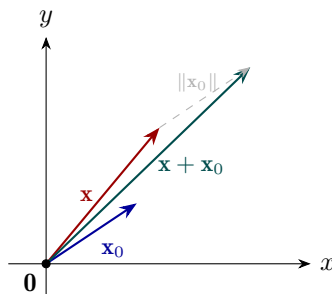


FIGURE 8.1.1 $T(\mathbf{x}) = \mathbf{x} + \mathbf{x}_0$ translates each point along a line parallel to $\mathbf{x}_0$ through distance $\|\mathbf{x}_0\|$.

EXAMPLE 9 | The Evaluation Transformation

Let V be a subspace of $F(-\infty, \infty)$, let $x_1, x_2, \dots, x_n$ be distinct real numbers, and define $T : V \rightarrow R^n$ by

$$T(f) = (f(x_1), f(x_2), \dots, f(x_n)) \quad (2)$$

This **evaluation transformation** is linear. For example, if $x_1 = -1$, $x_2 = 2$, $x_3 = 4$ and $f(x) = x^2 - 1$, then $T(f) = (f(-1), f(2), f(4)) = (0, 3, 15)$. Linearity follows from $T(kf) = k(f(x_1), \dots, f(x_n)) = kT(f)$ and $T(f + g) = T(f) + T(g)$.

Finding Linear Transformations from Images of Basis Vectors

For a matrix transformation $T_A : R^n \rightarrow R^m$ with standard basis $\mathbf{e}_1, \dots, \mathbf{e}_n$, any image $T_A(\mathbf{v}) = c_1 T_A(\mathbf{e}_1) + \dots + c_n T_A(\mathbf{e}_n)$ is determined by the images of the basis vectors. The following theorem extends this to general vector spaces.

Theorem 8.1.2

Let $T : V \rightarrow W$ be a linear transformation with V finite-dimensional. If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis for V , then for any $\mathbf{v}$ in V :

$$T(\mathbf{v}) = c_1 T(\mathbf{v}_1) + c_2 T(\mathbf{v}_2) + \dots + c_n T(\mathbf{v}_n) \quad (3)$$

where $c_1, c_2, \dots, c_n$ are the coordinates of $\mathbf{v}$ relative to S .

Proof. Write $\mathbf{v} = c_1 \mathbf{v}_1 + \dots + c_n \mathbf{v}_n$ and apply linearity of T . ■

EXAMPLE 10 | Computing with Images of Basis Vectors

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ for R^3 , where $\mathbf{v}_1 = (1, 1, 1)$, $\mathbf{v}_2 = (1, 1, 0)$, $\mathbf{v}_3 = (1, 0, 0)$. Let $T : R^3 \rightarrow R^2$ be linear with $T(\mathbf{v}_1) = (1, 0)$, $T(\mathbf{v}_2) = (2, -1)$, $T(\mathbf{v}_3) = (4, 3)$. Find a formula for $T(x_1, x_2, x_3)$, then compute $T(2, -3, 5)$.

SOLUTION: Write $(x_1, x_2, x_3) = c_1(1, 1, 1) + c_2(1, 1, 0) + c_3(1, 0, 0)$; solving gives $c_1 = x_3$, $c_2 = x_2 - x_3$, $c_3 = x_1 - x_2$. By Theorem 8.1.2:

$$\begin{aligned} T(x_1, x_2, x_3) &= x_3(1, 0) + (x_2 - x_3)(2, -1) + (x_1 - x_2)(4, 3) \\ &= (4x_1 - 2x_2 - x_3, 3x_1 - 4x_2 + x_3) \end{aligned}$$

Thus $T(2, -3, 5) = (4(2) - 2(-3) - 5, 3(2) - 4(-3) + 5) = (9, 23)$.

EXAMPLE 11 | A Linear Transformation from $C^1(-\infty, \infty)$ to $F(-\infty, \infty)$

(Calculus Required)

Let $V = C^1(-\infty, \infty)$ and $W = F(-\infty, \infty)$, and define $D : V \rightarrow W$ by $D(\mathbf{f}) = f'(x)$ (the differentiation operator). From calculus:

$$D(\mathbf{f} + \mathbf{g}) = D(\mathbf{f}) + D(\mathbf{g}) \quad \text{and} \quad D(k\mathbf{f}) = kD(\mathbf{f})$$

so D is a linear transformation.

EXAMPLE 12 | An Integral Transformation

(Calculus Required)

Let $V = C(-\infty, \infty)$, $W = C^1(-\infty, \infty)$, and define $J : V \rightarrow W$ by $J(f) = \int_0^x f(t) dt$. For example, if $f(x) = x^2$:

$$J(f) = \int_0^x t^2 dt = \left. \frac{t^3}{3} \right|_0^x = \frac{x^3}{3}$$

Linearity follows from properties of the integral: $J(kf) = k \int_0^x f = kJ(f)$ and $J(f + g) = \int_0^x (f + g) = J(f) + J(g)$.

Kernel and Range

For a matrix transformation T_A , the null space of A consists of all vectors mapped to $\mathbf{0}$, and the column space consists of all images. The following definition extends these concepts.

Definition 2

If $T : V \rightarrow W$ is a linear transformation, then the **kernel** of T is

$$\ker(T) = \{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{0}\}$$

and the **range** of T is

$$R(T) = \{\mathbf{w} \in W : \mathbf{w} = T(\mathbf{v}) \text{ for some } \mathbf{v} \in V\}$$

EXAMPLE 13 | Kernel and Range of a Matrix Transformation

If $T_A : R^n \rightarrow R^m$ is multiplication by $m \times n$ matrix A , then $\ker(T_A)$ is the null space of A and $R(T_A)$ is the column space of A .

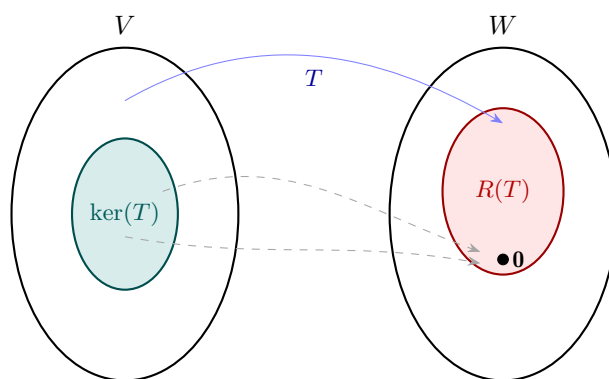


FIGURE 8.1.2 The kernel maps to $\mathbf{0}$; the range is the set of all images.

EXAMPLE 14 | Kernel and Range of the Zero Transformation

For the zero transformation $T : V \rightarrow W$: T maps every vector to $\mathbf{0}$, so $\ker(T) = V$. Since $\mathbf{0}$ is the only image, $R(T) = \{\mathbf{0}\}$.

EXAMPLE 15 | Kernel and Range of the Identity Operator

For $I : V \rightarrow V$: every vector is its own image so $R(I) = V$, and the only vector mapped to $\mathbf{0}$ is $\mathbf{0}$ itself, so $\ker(I) = \{\mathbf{0}\}$.

EXAMPLE 16 | Kernel and Range of an Orthogonal Projection

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the orthogonal projection onto the xy -plane: $T(x, y, z) = (x, y, 0)$. The points mapped to $\mathbf{0} = (0, 0, 0)$ are those of the form $(0, 0, z)$, so $\ker(T)$ is the z -axis. Every point in the xy -plane is the image of each point on the vertical line above it, so $R(T)$ is the entire xy -plane.

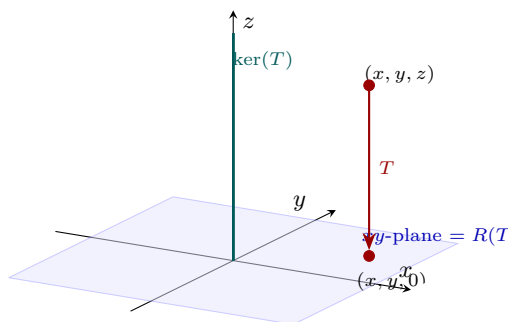


FIGURE 8.1.3 Projection onto the xy -plane: $\ker(T)$ is the z -axis and $R(T)$ is the xy -plane.

EXAMPLE 17 | Kernel and Range of a Rotation

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ rotate each vector through angle θ . Since every vector in the xy -plane can be obtained by rotating some vector through θ , we have $R(T) = \mathbb{R}^2$. The only vector that rotates into $\mathbf{0}$ is $\mathbf{0}$ itself, so $\ker(T) = \{\mathbf{0}\}$.

EXAMPLE 18 | Kernel of a Differentiation Transformation

(Calculus Required)

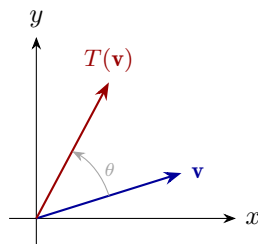


FIGURE 8.1.4 Rotation by θ : $\ker(T) = \{\mathbf{0}\}$, $R(T) = \mathbb{R}^2$.

For the differentiation operator $D : C^1(-\infty, \infty) \rightarrow F(-\infty, \infty)$ defined by $D(\mathbf{f}) = f'(x)$, the kernel consists of all functions with zero derivative, which is the set of constant functions on $(-\infty, \infty)$.

EXAMPLE 19 | Application to Differential Equations

(Calculus Required)

Differential equations of the form

$$y'' + \omega^2 y = 0 \quad (\omega \text{ a positive constant}) \quad (5)$$

arise in the study of vibrations. The set of all solutions on $(-\infty, \infty)$ is the kernel of $D : C^2(-\infty, \infty) \rightarrow C(-\infty, \infty)$ defined by $D(y) = y'' + \omega^2 y$. It can be shown that $y_1 = \cos \omega x$ and $y_2 = \sin \omega x$ are two linearly independent solutions of (5), so the kernel is a 2-dimensional subspace and every solution has the form

$$y = c_1 \cos \omega x + c_2 \sin \omega x \quad (6)$$

Properties of Kernel and Range

In all of the preceding examples $\ker(T)$ and $R(T)$ turned out to be subspaces. This is always the case.

Theorem 8.1.3

If $T : V \rightarrow W$ is a linear transformation, then:

- (a) The kernel of T is a subspace of V .
- (b) The range of T is a subspace of W .

Proof of (a). By Theorem 8.1.1(a), $\mathbf{0} \in \ker(T)$. If $\mathbf{v}_1, \mathbf{v}_2 \in \ker(T)$ and k is any scalar, then

$$T(\mathbf{v}_1 + \mathbf{v}_2) = T(\mathbf{v}_1) + T(\mathbf{v}_2) = \mathbf{0} + \mathbf{0} = \mathbf{0}$$

$$T(k\mathbf{v}_1) = kT(\mathbf{v}_1) = k\mathbf{0} = \mathbf{0}$$

so $\ker(T)$ is closed under addition and scalar multiplication. ■

Proof of (b). Since $T(\mathbf{0}) = \mathbf{0}$, we have $\mathbf{0} \in R(T)$. If $\mathbf{w}_1, \mathbf{w}_2 \in R(T)$ and k is any scalar, there exist $\mathbf{v}_1, \mathbf{v}_2$ in V with $T(\mathbf{v}_i) = \mathbf{w}_i$. Then

$$T(\mathbf{v}_1 + \mathbf{v}_2) = \mathbf{w}_1 + \mathbf{w}_2, \quad T(k\mathbf{v}_1) = k\mathbf{w}_1$$

so $\mathbf{w}_1 + \mathbf{w}_2 \in R(T)$ and $k\mathbf{w}_1 \in R(T)$. ■

Rank and Nullity of Linear Transformations

In Section 4.9 we defined rank and nullity for $m \times n$ matrices. These concepts extend to general linear transformations.

Definition 3

Let $T : V \rightarrow W$ be a linear transformation. If $R(T)$ is finite-dimensional, its dimension is called the **rank** of T . If $\ker(T)$ is finite-dimensional, its dimension is called the **nullity** of T . These are denoted

$$\text{rank}(T) \quad \text{and} \quad \text{nullity}(T)$$

Theorem 8.1.4 — Dimension Theorem for Linear Transformations

If $T : V \rightarrow W$ is a linear transformation from a finite-dimensional vector space V to a vector space W , then the range of T is finite-dimensional and

$$\text{rank}(T) + \text{nullity}(T) = \dim(V) \quad (7)$$

For the special case $T_A : R^n \rightarrow R^m$ (multiplication by A), $\ker(T_A)$ is the null space of A and $R(T_A)$ is the column space of A , so Theorem 8.1.4 gives the familiar result $\text{rank}(T_A) + \text{nullity}(T_A) = n$.

Exercise Set 8.1

In Exercises 1–2, suppose T is a mapping whose domain is the vector space M_{22} . Determine whether T is a linear transformation, and if so, find its kernel.

1. **a.** $T(A) = A^2$ **b.** $T(A) = \text{tr}(A)$ **c.** $T(A) = A + A^T$
3. $T : R^3 \rightarrow R$, where $T(\mathbf{u}) = \|\mathbf{u}\|$.
4. $T : R^3 \rightarrow R^3$, where $\mathbf{v}_0$ is a fixed vector in R^3 and $T(\mathbf{u}) = \mathbf{u} \times \mathbf{v}_0$.
5. $T : M_{22} \rightarrow M_{23}$, where B is a fixed 2×3 matrix and $T(A) = AB$.
7. $T : P_2 \rightarrow P_2$, where
 - a.** $T(a_0 + a_1x + a_2x^2) = a_0 + a_1(x+1) + a_2(x+1)^2$
 - b.** $T(a_0 + a_1x + a_2x^2) = (a_0+1) + (a_1+1)x + (a_2+1)x^2$

In Exercises 10–11, $T : P_2 \rightarrow P_3$ is defined by $T(p(x)) = xp(x)$.

10. Which of the following are in $\ker(T)$? **a.** x^2 **b.** 0 **c.** $1+x$ **d.** $-x$
11. Which of the following are in $R(T)$? **a.** $x+x^2$ **b.** $1+x$ **c.** $3-x^2$ **d.** $-x$
12. Let V be any vector space and $T : V \rightarrow V$ defined by $T(\mathbf{v}) = 3\mathbf{v}$.
 - a.** What is $\ker(T)$? **b.** What is $R(T)$?

13. Use the given information to find the nullity of T .
- a.** $T : R^5 \rightarrow P_5$ has rank 3. **b.** $T : P_4 \rightarrow P_3$ has rank 1. **c.** The range of $T : M_{mn} \rightarrow R^3$ is R^3 .
- d.** $T : M_{22} \rightarrow M_{22}$ has rank 3.
15. Let $T : M_{22} \rightarrow M_{22}$ be the dilation operator with factor $k = 3$.
- a.** Find $T\left(\begin{bmatrix} 1 & 2 \\ -4 & 3 \end{bmatrix}\right)$. **b.** Find the rank and nullity of T .
17. Let $T : P_2 \rightarrow R^3$ be the evaluation transformation at $x_1 = -1, x_2 = 0, x_3 = 1$.
- a.** Find $T(x^2)$. **b.** Find $\ker(T)$. **c.** Find $R(T)$.

In Exercises 19–22, consider the basis $S = \{\mathbf{v}_1, \mathbf{v}_2\}$ (or $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$) as given, and find a formula for $T(x_1, x_2)$ (or $T(x_1, x_2, x_3)$) using the stated images.

19. $S = \{\mathbf{v}_1, \mathbf{v}_2\}$ for R^2 : $\mathbf{v}_1 = (1, 1)$, $\mathbf{v}_2 = (1, 0)$; $T : R^2 \rightarrow R^2$ linear with $T(\mathbf{v}_1) = (1, -2)$, $T(\mathbf{v}_2) = (-4, 1)$. Find $T(5, -3)$.
21. $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ for R^3 : $\mathbf{v}_1 = (1, 1, 1)$, $\mathbf{v}_2 = (1, 1, 0)$, $\mathbf{v}_3 = (1, 0, 0)$; $T : R^3 \rightarrow R^3$ linear with $T(\mathbf{v}_1) = (2, -1, 4)$, $T(\mathbf{v}_2) = (3, 0, 1)$, $T(\mathbf{v}_3) = (-1, 5, 1)$. Find $T(x_1, x_2, x_3)$ and $T(2, 4, -1)$.

In Exercises 23–24, let $T_A : R^n \rightarrow R^3$ be multiplication by A . Find (a) a basis for the range of T , (b) a basis for the kernel of T , (c) the rank and nullity of T , (d) the rank and nullity of A .

23. $A = \begin{bmatrix} 1 & -1 & 3 \\ 5 & 6 & -4 \\ 7 & 4 & 2 \end{bmatrix}$

24. $A = \begin{bmatrix} 2 & 0 & -1 \\ 4 & 0 & -2 \\ 20 & 0 & 0 \end{bmatrix}$

In Exercises 27–28, determine whether T is linear, find a basis for the kernel, and find a basis for the range.

27. $T : P_3 \rightarrow P_2$ defined by $T(a_0 + a_1x + a_2x^2 + a_3x^3) = 5a_0 + a_3x^2$.
- a.** Show T is linear. **b.** Find a basis for $\ker(T)$. **c.** Find a basis for $R(T)$.
28. $T : P_2 \rightarrow P_2$ defined by $T(a_0 + a_1x + a_2x^2) = 3a_0 + a_1x + (a_0 + a_1)x^2$.
- a.** Show T is linear. **b.** Find a basis for $\ker(T)$. **c.** Find a basis for $R(T)$.
32. Let $T : M_{nn} \rightarrow R$ be defined by $T(A) = \text{tr}(A)$, where A is an $n \times n$ matrix with real entries. Determine the dimension of the range of T .
35. Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ be vectors in a vector space V , and let $T : V \rightarrow R^3$ be a linear transformation for which

$$T(\mathbf{v}_1) = (1, -1, 2), \quad T(\mathbf{v}_2) = (0, 3, 2), \quad T(\mathbf{v}_3) = (-3, 1, 2)$$

Find $T(2\mathbf{v}_1 - 3\mathbf{v}_2 + 4\mathbf{v}_3)$.

Working with Proofs

36. Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis for V and $T : V \rightarrow W$ a linear transformation. Prove: if $T(\mathbf{v}_1) = T(\mathbf{v}_2) = \dots = T(\mathbf{v}_n) = \mathbf{0}$, then T is the zero transformation.
38. Prove: if $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis for V and $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n$ are any vectors in W (not necessarily distinct), then there exists a unique linear transformation $T : V \rightarrow W$ such that $T(\mathbf{v}_k) = \mathbf{w}_k$ for $k = 1, 2, \dots, n$.